Summary Sheet

ProteinVision

SMILES Construction Rules

Quick reference for constructing and reading standard SMILES strings. Use our YouTube video for fuller explanations and examples.

1. Atoms and implicit hydrogens

- Write atoms using their element symbols. **Hydrogen atoms are normally omitted** when their number follows from the atom's normal valence and bonding.
- Example: `C` represents methane (CH4) because four hydrogens are implied.
- The following common elements may be written without square brackets when standard SMILES assumptions apply: `B, C, N, O, P, S, F, Cl, Br, I`.

Use square brackets when extra atom information must be stated explicitly:

- `[Fe]` - an atom outside the unbracketed organic subset.
- `[N-]` - a formal charge must be specified.
- `[OH-]` - an explicit hydrogen count and charge are required.
- `[13C]` - a non-default isotope is specified; no hydrogens are written here.
- `[13CH4]` - carbon-13 methane with four explicit hydrogens on the bracketed carbon.
- `[C@H]` / `[C@@H]` - stereochemistry around the atom is specified.

Practical rule: Use brackets if you need to state an isotope, formal charge, non-default hydrogen count, or atom stereochemistry explicitly.

2. Bonds

- **Single bond:** usually omitted; adjacent atom symbols imply a single bond. `CCO` and `C-C-O` describe the same connectivity.
- **Double bond:** `=` Example: `C=C`
- **Triple bond:** `#` Example: `C#N`

3. Branches

- Put a branch in **parentheses** immediately after the atom from which the branch starts.
- Example: `CC(=N)C`. The `(=N)` branch is attached to the second carbon.
- After a closing parenthesis, continue from the atom that opened the branch.
- Branches can be nested by placing parentheses inside parentheses.

4. Rings and aromatic atoms

- Mark a ring closure by writing the **same digit at both atoms** that are connected to close the ring.
- Example: `C1CCCCO1`. The two `1` labels identify the bond that closes the ring.
- Use different digits for different ring closures when several rings are open at the same time.
- For aromatic systems, write aromatic atoms with **lowercase symbols**; aromatic bond types are often left implicit.
- Example: benzene = `c1ccccc1`.

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5. Tetrahedral stereochemistry (chiral centers)

- Write the stereocenter in brackets and use `@` or `@@`.
- These symbols describe the arrangement **relative to the neighbor order in the SMILES string**; they do not directly mean R or S.

Interpretation rules:

1. Read the neighbors in the order encoded by the SMILES from left to right.
2. Look toward the stereocenter from the direction of the first neighboring atom.
3. For that neighbor order and viewing direction, `@` denotes anticlockwise and `@@` clockwise order of the remaining groups.

Example: `N[C@H](Cl)C` and `N[C@@H](Cl)C` encode opposite tetrahedral configurations when the rest of the SMILES is unchanged.

6. Double-bond stereochemistry (cis/trans)

- Use `/` and `\` on the bonds next to a double bond to specify relative geometry.
- In the simple linear example `F/C=C/F`, matching slashes place the fluorines on opposite sides: **trans**.
- In `F/C=C\F`, opposite slash directions place the fluorines on the same side: **cis**.
- The slash directions are relative to how the SMILES is written; interpret them as a pair around the double bond rather than as independent absolute directions.

7. Reaction SMILES

- Use a dot (.) to separate multiple molecules or components on the same side of a reaction.
- General format: reactants>agents>products.
- If no agents are specified, leave the middle field empty: reactants>>products.
- Agents can include reagents, catalysts, or other reaction agents.
- When explicit stoichiometric copies are included, repeat the molecule SMILES as needed.

Example - methane combustion: `C.O=O.O=O>>O=C=O.O.O`

8. SMILES are not unique

- The same molecular graph can often be written in more than one valid way because the traversal can start at different atoms.
- Example: `CCO` and `OCC` both represent ethanol.
- A **canonical SMILES** is a software-generated consistent representation. Canonical strings can differ between software packages because canonicalization algorithms can differ.

Construction checklist

1. Choose a path through the molecular graph and write the atoms in traversal order.
2. Omit ordinary hydrogens and single-bond symbols when they are implied.
3. Use parentheses for branches and matching digits for ring closures.
4. Add `=` or `#` for multiple bonds; use lowercase symbols for aromatic atoms.
5. Add square bracket information only when needed: isotope, charge, explicit H count, or tetrahedral stereochemistry.
6. Add `/` and `\` for double-bond stereochemistry.

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